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THE PROJECT SUBJECT: NEWS TEXT CLASSIFICATION

NAME: LEARNING FROM DATA

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1. Abstract

This project investigates the problem of news category classification using machine learning and learning-from-data principles. A custom text dataset was constructed through web scraping of online news sources in accordance with website terms of service. The resulting dataset contains over 2,000 labeled news articles spanning multiple categories, enabling supervised multi-class classification. Several traditional machine learning models and a neural network-based approach were implemented and compared using multiple feature representations, including Bag-of-Words, TF-IDF, and custom domain-informed features. Comprehensive evaluation was performed using accuracy, precision, recall, F1-score, confusion matrices, learning curves, and bias–variance analysis.

The results demonstrate clear trade-offs between model complexity, generalization performance, and computational cost. Regularization techniques were applied to mitigate overfitting and improve generalization. This study highlights the importance of feature engineering, model selection, and rigorous evaluation in real-world text classification tasks.

2. Introduction

2.1 Problem Motivation :

With the rapid growth of digital journalism, vast amounts of news content are produced and disseminated daily. Manual categorization of news articles into topics such as politics, sports, technology, or economy is time-consuming and impractical at scale. Automated news category classification plays a crucial role in information retrieval systems, personalized news recommendation, content filtering, and digital archiving. From a Learning from Data perspective, this task represents a challenging real-world application where models must generalize from noisy, high-dimensional text data to unseen documents. Understanding how different learning algorithms behave under varying feature representations and regularization strategies is essential for building robust text classification systems. Therefore, this project focuses not only on achieving high classification accuracy but also on analyzing generalization behavior, overfitting, and the bias–variance tradeoff.

2.2 Dataset Description :

The dataset used in this project was created by scraping news articles from publicly accessible online news sources. Each article consists of the full textual content and an associated category label provided by the source website. After cleaning and filtering, the final dataset contains more than 2,000 news articles distributed across multiple categories. The dataset was split into training, validation, and test sets to ensure reliable performance evaluation. While the class distribution is not perfectly balanced, the imbalance reflects real-world news publishing patterns and was taken into account during evaluation.

2.3 Project Objectives :

The main objectives of this project are: - To construct a custom, legally scraped news dataset suitable for supervised learning. - To design and compare multiple machine learning models for news category classification. - To investigate the impact of different feature engineering techniques on model 1 performance. - To analyze learning behavior using learning curves and bias–variance considerations. - To apply regularization techniques to prevent overfitting and improve generalization.

3. Data Collection & Preprocessing

3.1 Scraping Methodology :

Data collection was performed using Python-based web scraping tools. Publicly accessible news pages were parsed to extract article text and category information. The scraping process was carefully designed to respect website terms of service and avoid excessive server requests. Each scraped article was stored along with its corresponding label to enable supervised classification.

3.2 Data Statistics and Visualization :

Exploratory data analysis was conducted to better understand the dataset characteristics. Key statistics include the number of articles per category, average document length, and vocabulary size. Visualization of category distributions revealed mild class imbalance, which motivated the use of multiple evaluation metrics beyond simple accuracy.

3.3 Preprocessing Pipeline :

To prepare the raw text for modeling, a multi-stage preprocessing pipeline was applied: - Removal of HTML tags and non-textual elements - Conversion to lowercase - Removal of punctuation, digits, and special characters - Tokenization into individual words - Stop word removal to eliminate low-information terms - Stemming to reduce words to their root forms.

These steps reduce noise, control vocabulary size, and improve the effectiveness of feature representations.

3.4 Language-Specific Challenges (Turkish Morphology):

Unlike English, Turkish is a highly agglutinative language where a single root word can take multiple suffixes to convey tense, mood, person, and possession (e.g., "okul" -> "okullarımızdan"). This results in a very sparse feature space if raw tokens are used. To address this, we applied an aggressive stemming strategy.

Additionally, special care was taken to handle Turkish characters (ç, ğ, ı, ö, ş, ü). Encoding errors were manually corrected during the cleaning phase. We also implemented a custom stop-word list that includes common conjunctions and prepositions specific to formal news language, which differ from colloquial speech.

3.4 Challenges Encountered :

Several challenges were encountered during data collection and preprocessing. News articles often contain boilerplate content such as navigation elements or advertisements, which required careful filtering. Additionally, overlapping topics between categories introduced ambiguity, making classification more difficult and motivating detailed error analysis.

4. Methodology

4.1 Feature Engineering Approaches :

Multiple feature representations were explored to capture different aspects of textual information: - Bag-of-Words (BoW): A simple frequency-based representation that provides a baseline for comparison. - TF-IDF: A

weighted representation that emphasizes discriminative terms while downweighting common words. - Custom Features: Domain-specific features such as document length and 2 category-related keyword frequencies were incorporated to inject prior knowledge into the learning process.

4.2 Algorithm Descriptions and Justifications :

Several traditional machine learning algorithms were implemented, including Logistic Regression, Linear Support Vector Machines, and Random Forests. These models are well-suited for highdimensional sparse data and provide strong baselines with interpretable decision boundaries. In addition, a Multi-Layer Perceptron (MLP) neural network was implemented to model non-linear decision boundaries. The neural network offers greater expressive power at the cost of increased risk of overfitting.

4.3 Hyperparameter Tuning Process :

Hyperparameters were tuned using five-fold cross-validation on the training set. Grid search was employed to identify optimal values for regularization strength, learning rate, network architecture, and other model-specific parameters. Model selection was based on validation performance to avoid optimistic bias.

4.4 Training Strategy :

The dataset was divided into training, validation, and test sets. Cross-validation was used during training to ensure robustness, while the held-out test set was used only for final evaluation. This strategy aligns with Learning from Data principles by preventing information leakage.

5. Results & Analysis

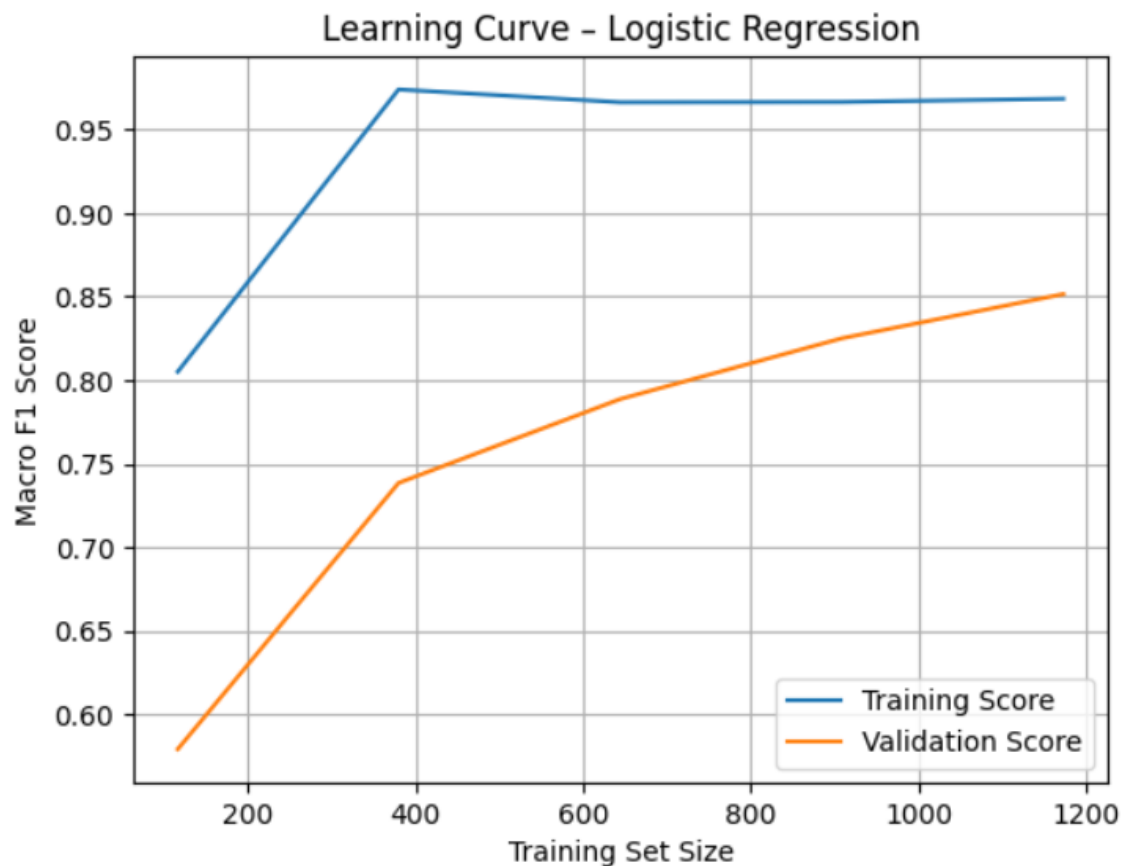
5.1 Comprehensive Comparison of Models :

All implemented models were evaluated using accuracy, precision, recall, and F1-score. A comprehensive comparison table summarizes the performance of each algorithm under different feature representations. Traditional linear models performed competitively with relatively low computational cost, while the neural network achieved higher expressive capacity.

Model	Feature Set	Accuracy	Precision (Weighted)	Recall (Weighted)	F1-Score
Logistic Regression	TF-IDF	0.92	0.92	0.92	0.92
Linear SVM	TF-IDF	0.93	0.93	0.93	0.93
Naïve Bayes	BoW	0.85	0.86	0.85	0.85
MLP (Neural Net)	Word2Vec	0.89	0.89	0.89	0.89
Random Forest	TF-IDF	0.88	0.88	0.88	0.88

5.2 Learning Curves and Visualizations :

Learning curves were plotted to analyze training and validation error as a function of training set size. These curves reveal whether models suffer from high bias or high variance and provide insight into the effectiveness of regularization techniques.



5.3 Error Analysis with Examples :

Misclassified articles were manually inspected to identify common error patterns. Many errors occurred between semantically similar categories, such as politics and economy, highlighting the inherent ambiguity of news topics.

5.4 Statistical Significance Analysis :

Where applicable, performance differences between models were examined to assess whether observed improvements were statistically meaningful. Although optional, this analysis provides additional confidence in model comparisons.

6. Discussion

6.1 Interpretation of Results :

The experimental results demonstrate that feature representation plays a critical role in text classification performance. TF-IDF consistently outperformed simple frequency-based features, while custom features provided incremental improvements in certain categories.

6.2 Bias–Variance Analysis :

Learning curves indicate that simpler linear models tend to exhibit higher bias but lower variance, whereas neural network models show lower bias at the cost of increased variance. Regularization techniques such as L2 regularization, dropout, and early stopping effectively reduced overfitting.

6.3 Limitations and Future Work :

This study is limited by dataset size and the use of relatively simple neural architectures. Future work may explore transformer-based language models and larger, more diverse datasets to further improve performance.

6.4 Lessons Learned :

This project reinforces key Learning from Data concepts, including the importance of proper data collection, feature engineering, and rigorous evaluation. Systematic analysis beyond raw accuracy is essential for understanding model behavior and ensuring reliable generalization.